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# SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

**Table 1: Rigor Adherence Table**

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|---|
| Ethics |
| not detected. |
| Inclusion and Exclusion Criteria |
| Search criteria included carbamidomethylation of cysteine ( +57.02 Da ) as a fixed modification and oxidation of methionine ( +15.99 Da ) as a variable modification . |
| Attrition |
| not required. |
| Sex as a biological variable |
| not detected. |
| Subject Demographics |
| Age: not detected. |
| Weight: not detected. |
| Randomization |
| not detected. |
| Blinding |
| not detected. |
| Power Analysis |
| not detected. |
| Replication |
| Number: RT-PCR analysis of RNA samples To detect the sRNAs under study , input , affinity purified and CoIP RNA from 24 OD600 equivalents ( 1 ml ) ( at least two independent eluates per sample ) were subjected to First-Strand cDNA Synthesis Using 100 U SuperScript(tm ) II RT ( Invitrogen) . |
| Data Information |

| | |
|---|---|
| Availability: The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE 71 partner repository with the dataset identifier PXD006410 . | |
| Identifiers: The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE 71 partner repository with the dataset identifier PXD006410 . | PXD006410 |
| Identifiers: CoIP assays with 3x FLAG-tagged proteins For co-immunoprecipitation ( CoIP) , S. meliloti carrying the plasmids encoding FLAG-tagged proteins were induced with IPTG and harvested in the conditions resembling cognate sRNA expression described before 29,30,33 . | 29,30,33 |

**Table 2: Key Resources Table**

| Your Sentences | REAGENT or RESOURCE | SOURCE | IDENTIFIER |
|---|---|---|---|
| Antibodies | | | |
| Membranes were probed with a monoclonal anti-FLAG antibody ( Sigma F7425 1:5,000 ) as reported 67 . | anti-FLAG | | |
| Experimental Models: Organisms/Strains | | | |
| Plasmids pSKEcpR1+ and pSKAbcR2+ were generated by PCR amplification of EcpR1 from its prevalent processed 5-end and full-length AbcR2 , respectively , as described 29,30 . Strain SmmetKFLAG , the Sm2B2019 derivative chromosomally encoding a MetK variant tagged with the FLAG epitope at the C-terminus , was generated by replacement of the wild-type allele via pK18metKFLAG -mediated double recombination as described 67 | Sm2B2019 | | |
| Recombinant DNA | | | |
| For systematic 5-end aptamer-tagging of the sRNAs of interest , MS2 and MS2* aptamers were generated by annealing of two complementary oligonucleotides that leave BamHI and XbaI compatible overhangs and then inserted between these sites into the mid-copy pSRKKm derivative pSRK-C 29,65 , yielding vectors pSRK-MS2 and pSRK-MS2* . | pSRK-C 29,65 | | |
| These PCR products were digested and inserted between the corresponding sites in pSRK-MS2 and pSRK-MS2* to yield pSRK-MS2MS2*-AbcR2 , pSRK-MS2MS2*-EcpR1 and pSRK-MS2MS2*-NfeR1 that express the MS2MS2* 5-tagged sRNAs constitutively . | pSRK-MS2 | | |
| | pSRK-MS2* | | |
| | pSRK-MS2MS2*-AbcR2 | | |
| | pSRK-MS2MS2*-EcpR1 | | |
| | pSRK-MS2MS2*-NfeR1 | | |
| A Rho-independent transcriptional terminator ( T1 ) generated by oligo hybridization was also cloned downstream of MS2 and MS2* to | pSRK-MS2MS2*-Term | | |

| | | | |
|---|---|---|---|
| generate the control plasmids pSRK-MS2MS2*-Term . | | | |
| Control plasmid pSRK-EcpR1 constitutively overexpressing the untagged wild-type EcpR1 sRNA from its transcription starting site 2 ( TSS2 ) 30 was similarly obtained by PCR amplification and cloning into pSRK-C as a BamHI- SacI fragment . | pSRK-EcpR1 | | |
| | pSRK-C | | |
| To construct vector pSK-FLAG , the complementary oligos XBTagF and BXTagR , carrying the 3x FLAG sequence followed by a TGA stop codon , were annealed and cloned as XbaI-BamHI fragments into pSRKKm . | pSRKKm | | |
| The coding regions of selected proteins , devoid of the respective stop codon , were amplified by PCR using primers that introduce NdeI and XbaI sequences to be inserted into the corresponding restriction sites of pSK-FLAG . | pSK-FLAG | | |
| Plasmids pSKEcpR1+ , pSKNfeR1+ 33 , and pSKAbcR2+ , were similarly mobilized as required to strains Sm4011ecpR1 30 , SmmetKFLAG or Sm2B2020 for IPTG-induced transcription of the untagged wild-type EcpR1 , NfeR1 and AbcR2 sRNAs , respectively . | pSKAbcR2+ | | |
| Plasmids pSKEcpR1+ and pSKAbcR2+ were generated by PCR amplification of EcpR1 from its prevalent processed 5-end and full-length AbcR2 , respectively , as described 29,30 . Strain SmmetKFLAG , the Sm2B2019 derivative chromosomally encoding a MetK variant tagged with the FLAG epitope at the C-terminus , was generated by replacement of the wild-type allele via pK18metKFLAG -mediated double recombination as described 67 | pSKEcpR1+ | | |
| Plasmid pK18metKFLAG was generated by PCR amplification of the MetK-FLAG coding sequence from pSK_MetKFLAG and a 1,300-bp fragment of its downstream region from Sm2B3001 genomic DNA with primer pairs EcoNdemetkFBamTGAflagR and 3metKOBamHI_F4metKOPstII_R , respectively . | pK18metKFLAG | | |
| | pSK_MetKFLAG | | |

| | | | |
|---|---|---|---|
| In the same ligation reaction , both digestion products were inserted between the EcoRI and XbaI sites of the suicide vector pK18mobsacB 68 to yield pK18metK FLAG . | pK18mobsacB | | |
| To achieve MetK depletion in strain Sm2011 , its coding sequence was deleted by pK18DmetK-mediated double recombination as described 29 in the presence of plasmid pSKmetK , which expresses the wild-type metK gene upon IPTG induction . | pSKmetK | | |
| Plasmid pK18DmetK was generated by PCR amplification of Sm2011 DNA with the pair of primers 1metKOEcoRI_F2metKKOBamHI_R and 3metKOBamHI_F4metKOPstII_R , ligation of the resulting fragments at cohesive BamHI overhangs and insertion of the tandem between the EcoRI and PstII sites of pK18mobsacB . | pK18DmetK | | |
| Finally , as reporter of intracellular SAM accumulation we constructed plasmid pRmetZ::eGFP , which expresses a translational fusion of the SAM-II riboswitch , preceding the metZ gene , to eGFP . | pRmetZ::eGFP | | |
| Fluorescence of S. meliloti single or double transconjugants carrying the reporter construct pR_eGFP was measured on adjusted stationary TY cultures in a Tecan Infinite M200 reader . | pR_eGFP | | |
| The fragment was checked by sequencing and inserted between the NdeI and XhoI sites of pET-16b ( Novagen) , yielding p16metK , which encodes a MetK variant tagged with 10 His residues at its N-terminus . | pET-16b | | |
| For purification of recombinant MetK , plasmid p16metK was transformed into Rosetta-gami ( DE3 ) pLysS ( Novagen) . | p16metK | | |
| In vitro synthesis and labelling of RNA The AbcR2 , NfeR1 , EcpR1 , SmelC812 38 , tRNAMet , SMb20420 and 5S rRNA transcripts were amplified for in vitro transcription from Sm2B3001 genomic DNA and the group II ribozyme RmInt1 from plasmid pKGEMA4 42 with the primer pairs listed in Supplementary Table S4 . | pKGEMA4 | | |

| Software and Algorithms | | | |
|---|---|---|---|
| The raw data files were processed using DataAnalysis software v4.3 ( Bruker-Daltonics ) and searched against the UniProtTrembl database using Mascot 2.4 ( Matrix Science ) integrated together with ProteinScape v4.0 ( Bruker-Daltonics) | ProteinScape | | |
| The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE 71 partner repository with the dataset identifier PXD006410 . | PRIDE | | Suggestion: Pride-asap (RRID:SCR_012052)( link) |
| Reactions were loaded into a dot-blot device (Bio-Dot Bio-Rad) provided with nitrocellulose and PVDF membranes (Amersham). | Bio-Dot | | |
| Membranes were finally exposed to a Phosphor Imager screen (Bio-Rand) and quantified with the Quantity One software (Rio-Rad). | Quantity One | | |

**Other Entities Detected**

| Your Sentences | Recognized Entity |
|---|---|
| Statistical Tests | |
| Relationship between samples was assessed using the number of peptides to calculate the Pearson correlation coefficient in excel ( r) . | Pearson correlation |

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For a full description of scored criteria and tips for improving your score, please see https://www.scicrunch.com/sciscorereport-faq